

ARYAN DHANUKA

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EDUCATION

Bennett University

Greater Noida, India

Graduation July 2027

B.Tech Computer Science Engineering (AI Specialization) - CGPA: 8.84

TECHNICAL SKILLS

Languages: Python, C++

ML/AI Frameworks: TensorFlow, Scikit-learn, XGBoost, Pandas, NumPy, HuggingFace (Transformers/Diffusers), LangChain, LlamaIndex, OpenCV, NLP

Backend & Deployment: FastAPI, Flask, Streamlit, Docker, AWS (ECR, EC2), Vercel, Streamlit Cloud

Databases & Retrieval: Pinecone, FAISS

LLM & APIs: OpenAI GPT-4o, ChatGroq (Llama-3.3-70b), HuggingFace Hub

DevOps & Tools: Linux, Docker Compose, CI/CD Pipelines, Git

EXPERIENCE

Open Source Contributor — sktime

Oct 2025

- Developed scenario-based tests for detector estimators and implemented DetectorUnivariateSimple with comprehensive smoke tests to improve reliability in the sktime production repository[#8914]
- Enhanced TimeBinner transformer with advanced test parameter sets, increasing robustness for time-series binning operations and addressing critical edge-case failures in the sktime production repository[#8865]

AI Engineering Intern — LearnNex (Powered by Wipro)

2025

- Built a production-grade multi-domain LLM system with domain-aware routing and modular agents.
- Developed a hybrid RAG pipeline (BM25 + dense embeddings) for grounded, context-aware responses.
- Engineered a document intelligence system with PDF ingestion, chunking, and Map-Reduce summarization.
- Deployed scalable FastAPI inference APIs with streaming and low-latency response handling.
- Designed full-stack architecture using React (Vite) and integrated LLM + embedding workflows.

PROJECTS

AI Manufacturing Quality Failure Detection

PyTorch — FastAPI — Docker — Hugging Face

[Github Repository](#)

- Engineered a hybrid multimodal system fusing CNN-based visual defect detectors with Isolation Forest sensor anomaly models, validated on MVTec and NASA Turbofan datasets.
- Developed a calibrated risk-fusion layer and uncertainty-aware HITL (Human-in-the-Loop) mechanism to flag ambiguous predictions, reducing false negatives in safety-critical workflows.
- Built a real-time telemetry dashboard using Next.js and Recharts, integrated with a FastAPI backend achieving sub-100ms inference through optimized model caching.
- Implemented modular inference pipelines with Pydantic validation, ensuring system robustness against noisy or missing multimodal sensor inputs.

Explainable Multimodal Credit Risk Platform

LightGBM — PyTorch — DVC — Docker — Scikit-learn

[Github Repository](#)

- Architected a tri-modal risk engine combining tabular credit data, time-series behavioral features, and ResNet18-based document quality embeddings for fraud detection.
- Designed leakage-safe feature engineering pipelines using entity-aware rolling windows and DVC for 100% experiment reproducibility and data versioning.
- Integrated synthetic data augmentation (blur, noise, cropping) to train defensive ML models capable of identifying corrupted or fraudulent financial documentation.
- Built automated unit testing with Pytest for feature leakage checks and class-imbalance handling, ensuring stable deployment on Hugging Face Spaces.

AI Chatbot Using LLMs

NLP — FastAPI — React Vite — OpenAI GPT-4o — Docker

[Github Repository](#)

- Engineered 5-domain router with confidence thresholds achieving 100% uptime with zero crashes on ambiguous inputs.
- Implemented WebSocket streaming reducing TTFT to less than 500ms with Groq, 800ms with GPT-4o.
- Optimized FastAPI singleton with LRU caching reducing cold start from 2.5s to less than 500ms.
- Replaced OpenAI embeddings with local HuggingFace MiniLM-L6-v2, eliminating per-token costs and reducing latency by 200ms.

COMPETITIVE PROGRAMMING

Codolio: [AryanD10](#)